



MODUL **PRAKTIKUM**

ADMINISTRASI BASIS DATA

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Materi 8

Transaction Log Backup & Point-in-Time Recovery

A. Tujuan

Setelah mempelajari materi ini, mahasiswa diharapkan mampu:

- Mengaktifkan FULL Recovery Model pada DBKampus
- Melakukan Full Backup sebagai dasar log
- Melakukan Transaction Log Backup berurutan
- Mensimulasikan human error
- Melakukan Point-in-Time Recovery
- Menganalisis kegagalan restore log

B. Persiapan

- Database: DBKampus sudah ada
- Folder backup tersedia

C. Praktikum

LANGKAH 1 — CEK & UBAH RECOVERY MODEL

Cek recovery model:

```
SELECT name, recovery_model_desc
FROM sys.databases
WHERE name = 'DBKampus';
```

70 %

Results Messages

	name	recovery_model_desc
1	DBKampus	FULL

Jika masih SIMPLE, ubah:

```
ALTER DATABASE DBKampus
SET RECOVERY FULL;
GO
```

Penting:

Setelah mengubah ke FULL, WAJIB melakukan FULL BACKUP terlebih dahulu sebelum log backup bisa dilakukan.

LANGKAH 2 — FULL BACKUP (DASAR LOG)

```
BACKUP DATABASE DBKampus
TO DISK = 'D:\BackupSQL\DBKampus_Full12.bak'
WITH INIT,
    STATS = 10;
GO
```

70 %

Messages

11 percent processed.
20 percent processed.
30 percent processed.
40 percent processed.
51 percent processed.
61 percent processed.
70 percent processed.
80 percent processed.
90 percent processed.
100 percent processed.
Processed 512 pages for database 'DBKampus', file 'DBKampus' on file 1.
Processed 1 pages for database 'DBKampus', file 'DBKampus_log' on file 1.
BACKUP DATABASE successfully processed 513 pages in 0.049 seconds (81.662 MB/sec).
Completion time: 2026-05-05T05:50:30.5693814+07:00

LANGKAH 3 — SIMULASI TRANSAKSI

Gunakan tabel yang sudah ada, misalnya:

- Mahasiswa

Tambah Data

```
INSERT INTO akademik.Mahasiswa (NIM, Nama, TanggalLahir, Prodi, TahunMasuk)
VALUES ('999', 'Simulasi1', '2004-04-11', 'Teknologi Informasi', '2025');
```

70 %

Messages

(1 row affected)
Completion time: 2026-05-05T05:57:04.7500157+07:00

LANGKAH 4 — LOG BACKUP PERTAMA

```
BACKUP LOG DBKampus
TO DISK = 'D:\BackupSQL\DBKampus_Log1.trn'
WITH INIT,
    STATS = 10;
GO
```

70 %

Messages

61 percent processed.
100 percent processed.
Processed 13 pages for database 'DBKampus', file 'DBKampus_log' on file 1.
BACKUP LOG successfully processed 13 pages in 0.004 seconds (25.268 MB/sec).
Completion time: 2026-05-05T05:58:21.9190466+07:00

Log1 berisi perubahan setelah Full Backup.

LANGKAH 5 — TRANSAKSI KEDUA

```
INSERT INTO akademik.Mahasiswa (NIM, Nama, TanggalLahir, Prodi, TahunMasuk)
VALUES ('998', 'Simulasi2', '2004-04-11', 'Teknologi Informasi', '2025');
```

70 %

Messages

(1 row affected)

Completion time: 2026-05-05T06:00:43.6998092+07:00

LANGKAH 6 — LOG BACKUP KEDUA

```
BACKUP LOG DBKampus
TO DISK = 'D:\BackupSQL\DBKampus_Log2.trn'
WITH INIT,
STATS = 10;
GO
```

70 %

Messages

100 percent processed.
Processed 2 pages for database 'DBKampus', file 'DBKampus_log' on file 1.
BACKUP LOG successfully processed 2 pages in 0.003 seconds (3.580 MB/sec).

Completion time: 2026-05-05T06:01:55.8698584+07:00

LANGKAH 7 — SIMULASI HUMAN ERROR

Jalankan:

```
DELETE FROM Akademik.Mahasiswa;
GO
```

70 %

Messages

(6 rows affected)

Completion time: 2026-05-05T06:04:05.3168614+07:00

Cek:

```
SELECT * FROM Akademik.Mahasiswa;
```

70 %

Results Messages

NIM	Nama	TanggalLahir	Prodi	TahunMasuk
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Data kosong.

Catat waktu eksekusi DELETE (misal: 14:35:00).

LANGKAH 8 — RESTORE POINT-IN-TIME

Sekarang kita akan mengembalikan database sebelum DELETE.

1. Restore Full Backup

```
/*Master*/
RESTORE DATABASE DBKampus
FROM DISK = 'D:\BackupSQL\DBKampus_Full2.bak'
WITH NORECOVERY,
REPLACE;
GO
```

70 %

Messages

Processed 512 pages for database 'DBKampus', file 'DBKampus' on file 1.
Processed 1 pages for database 'DBKampus', file 'DBKampus_log' on file 1.
RESTORE DATABASE successfully processed 513 pages in 0.037 seconds (108.147 MB/sec).
Completion time: 2026-05-05T06:09:18.5698674+07:00

2. Restore Log1

```
RESTORE LOG DBKampus
FROM DISK = 'D:\BackupSQL\DBKampus_Log1.trn'
WITH NORECOVERY;
GO
```

70 %

Messages

Processed 0 pages for database 'DBKampus', file 'DBKampus' on file 1.
Processed 13 pages for database 'DBKampus', file 'DBKampus_log' on file 1.
RESTORE LOG successfully processed 13 pages in 0.007 seconds (14.439 MB/sec).
Completion time: 2026-05-05T06:11:36.8898377+07:00

3. Restore Log2 dengan STOPAT

Misalnya DELETE terjadi pukul:

2026-05-05 06:10:00

Maka:

```
RESTORE LOG DBKampus
FROM DISK = 'D:\BackupSQL\DBKampus_Log2.trn'
WITH STOPAT = '2026-05-05 06:10:59',
RECOVERY;
GO
```

70 %

Messages

Processed 0 pages for database 'DBKampus', file 'DBKampus' on file 1.
Processed 2 pages for database 'DBKampus', file 'DBKampus_log' on file 1.
This backup set contains records that were logged before the designated point in time. The database is being left in the restoring state so that more roll forward can be performed.
RESTORE LOG successfully processed 2 pages in 0.008 seconds (1.342 MB/sec).
Completion time: 2026-05-05T06:19:01.4449671+07:00

LANGKAH 9 - VERIFIKASI

1. Cek status database

```
SELECT name, state_desc
FROM sys.databases
WHERE name = 'DBKampus';
```

70 %

Results Messages

	name	state_desc
1	DBKampus	RESTORING

2. Cara Mengecek Progress Restore

```
SELECT
    percent_complete,
    start_time,
    estimated_completion_time/1000 AS estimasi_detik,
    command
FROM sys.dm_exec_requests
WHERE command LIKE 'RESTORE%';
```

Query tersebut akan menampilkan:

- Persentase progress
- Estimasi waktu selesai

tidak menampilkan data apa pun, artinya:

- Restore sudah selesai
- Restore sudah gagal
- Restore tidak pernah berjalan
- Atau database sudah dalam status RESTORING tapi tidak ada proses aktif

3. Solusi Jika Restore Terlalu Lama

- Status RESTORING (Tapi Tidak Ada Progress)

Artinya restore sebelumnya dijalankan dengan:

WITH NORECOVERY

Dan belum ditutup dengan:

RESTORE DATABASE DBKampus WITH RECOVERY;

Solusi:

RESTORE DATABASE DBKampus WITH RECOVERY;

70 %

Messages

DBCC execution completed. If DBCC printed error messages, contact your system administrator.
 RESTORE DATABASE successfully processed 0 pages in 0.972 seconds (0.000 MB/sec).

Completion time: 2026-05-05T06:29:11.7661124+07:00

4. Cek ulang isi tabel

SELECT*FROM akademik.Mahasiswa

70 %

Results **Messages**

	NIM	Nama	TanggalLahir	Prodi	TahunMasuk
1	20250140001	Andi	2004-02-10	Teknologi Informasi	2025
2	20250140002	Siti	2003-05-12	Teknologi Elektro	2025
3	20250140003	Budi	2004-08-15	Teknologi Informasi	2025
4	20250140004	Dewi	2003-11-21	Teknologi Elektro	2025
5	998	Simulasi2	2004-04-11	Teknologi Informasi	2025
6	999	Simulasi1	2004-04-11	Teknologi Informasi	2025